

Sari-Sari Store Simulator

A Python-Based Retail Business Simulation and Analytics Tool

DSC 512 project by Learning Team 6 (Jaco Abat, Lean Jerios, John Lagac, Justin Rotairo)

Context:

This project is a standalone three-level Python program that simulates the operation of a small sari-sari store. The project starts with a basic sales simulator, evolves into a month-long store management game, and finally becomes an analytics-driven decision-support system that uses past sales data to recommend inventory and pricing decisions.

Level	Standalone Program	Main Focus	How to Test
Basic	One-day Sari-Sari store sales calculator - Calculate the revenues and expenses extracted from a file of <u>a day's worth of transactions</u> - Track remaining stocks for the day	Selling, profit	Full ledger check and balance trace. All stocks tally, revenue and expenses adds up
Intermediate	One-month Sari-Sari store sales simulator with Analytics - Calculate the revenues and expenses extracted from a file of <u>a month's worth of transactions</u> - Create file/s of inventory for the month prior to sales and another file of inventory needed to restock for the next month - Create a function that generates a monthly dashboard and analytics report - Create a function that automatically generates synthetic data of a month's worth of transactions - Function should also accept benchmark information about the Sari-Sari store industry in the PH for the mock data to be representative of real-life conditions	Restocking, dashboard and analytics report, Synthetic data generation	Comparison of revenues generated to real data, Accuracy of information reflected in Dashboard and Inventory
Advanced	Inventory Optimization Engine and Pricing Strategy Analysis - Calculate the revenues and expenses extracted from a file of <u>5 years worth of transactions</u> - Create a function that automatically generates 5 years worth of transactions (1 folder per year, 1 folder per month) - On a per month basis: - Create a function that automatically generates synthetic data of a month's worth of transactions - For monthly transactions, add occurrences of grocery-wide sales - Create file/s of inventory for the month prior to sales and another file of inventory needed to restock for the next month - Create a function that generates customer reviews/feedback for the month - Create an Inventory Optimization engine/Pricing Strategy Analysis that does the ff: - Recommend which stocks should be restocked - Recommend the price to sell these stocks based on historical data	Inventory Optimization, Pricing Recommendation	Comparison of Recommended Prices for select goods to real-life prices of competitors, Dashboard to reflect peak in sales when adding occurrences of grocery-wide sales
Legends <u>Enhancement from previous level</u> - New function added for level			